

Computer Science undergraduate and software engineer building research platforms, production web applications, and reliable data workflows. Experience across digital health, full-stack development, REST APIs, authentication, databases, machine-learning experiments, and startup product delivery.

EDUCATION

Punjabi University, Patiala — B.Tech., Computer Science & Engineering | CGPA: 7.89/10

2023–2027

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Artificial Intelligence, Machine Learning

RESEARCH & EXPERIENCE

IIT Ropar — Summer Research Intern

May–Jul 2026

Digital Health Research Software · Supervisor: Dr. Sujata Pal

- Engineered **two full-stack digital-health research platforms** spanning participant applications, longitudinal study workflows, researcher dashboards, and wearable-data integration.
- Built Node.js/Express REST APIs, MongoDB/Mongoose data models, JWT authentication, email verification, and role-based access control; validated API contracts, form data, permissions, and dashboard outputs.
- Migrated a study backend from FastAPI/PostgreSQL to Node.js/MongoDB while preserving frontend API compatibility and research workflows.

Grubox — Software Development & Marketing Intern

Jan 2025–Present

Migrated the company's production website from Wix to **Next.js** and continue delivering frontend features, content systems, deployment updates, and digital growth initiatives.

BuildVR — Software Development & Marketing Intern

Jan 2025–Present

Develop multilingual landing pages and support immersive VR/3D product experiences through UI implementation, testing, 3D assets, deployment, and AI-assisted content workflows.

Punjabi University — Academic Research

Aug–Dec 2024

Reviewed and synthesized research on neural-network architectures for 5G/6G communication systems under Snehkunwar Sidhu.

SELECTED PROJECTS

Women's Health Research Platform ■

React · TypeScript · Node.js · Express · MongoDB · JWT

Longitudinal research system for symptom and menstrual tracking, questionnaires, physiological data collection, and researcher review.

Symptom Scope ■

Flutter · Dart · Offline-First Mobile

Privacy-first mobile triage app with adaptive follow-ups, red-flag screening, condition ranking, local history, and shareable reports.

Stress Management Research Platform ■

React · TypeScript · REST APIs · MongoDB · JWT

Session-based platform connecting validated stress assessments and wearable measurements with authenticated researcher dashboards.

When Systems Break ■

Python · Pandas · NumPy · Matplotlib · ML Reliability

Experimental framework measuring model behaviour under missing data, noise, feature degradation, and distribution shift.

TECHNICAL SKILLS

Languages

Python, C++, C, JavaScript, TypeScript, SQL, Dart

Web & Backend

React, Next.js, Node.js, Express.js, Vite, REST APIs, JWT, RBAC, Flutter

Data & ML

MongoDB, Mongoose, PostgreSQL, Pandas, NumPy, Matplotlib, Scikit-learn, SHAP

Developer Tools

Git, GitHub, Linux, VS Code, Postman, API testing, deployment

TRAINING

IIT Kanpur — Full-Stack Web Development, MERN · Jun–Jul 2024

ISRO / IIRS — AI/ML for Geodata Analysis · Aug–Sep 2024

GDGoC PUP — AI/ML Express · **Top Achiever**

LEADERSHIP & ACHIEVEMENTS

Winner, GDG TechSprint Hackathon · Contributor, GSSoC 2025

Executive Member, Technical Society · **150+ LeetCode** problems

National-Level Gold Medalist, Avantika Painting Competition